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MILITARY SPECIFICATION

PUMP UNIT, CENTRIFUGAL, VERTICAL SUMP, ELECTRIC MOTOR DRIVEN, AUTOMATIC, WET-PIT-TYPE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers automatic, wet-pit-type, electric motor driven, centrifugal, vertical sump pumps for handling drainage and waste water.

1.2 Classification. Pump units shall be of the following capacity groups and total dynamic heads as specified (see 3.7 and 6.2):

Group I - 5 gallons per minute (gpm) to 75 gpm.
Group II - above 75 gpm to 600 gpm.

Total dynamic head - 10 feet to 120 feet of water.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (for the Treatment of Communications, Electronics and Associated Electrical Equipment).

MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power Driven, for Water, Packaging of.

STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government document. The following other Government document form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A120 - Specification for Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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HYDRAULIC INSTITUTE (HI)

Standards of Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, 14600 Detroit Avenue, Cleveland, OH 44107.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS-1 - General Standards for Industrial Controls and Systems.
ICS-2 - Industrial Control Devices, Controllers and Assemblies.
ICS-6 - Enclosures for Industrial Controls and Systems.
MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITERS LABORATORIES, INC. (UL)

Electrical Construction Materials List.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. Centrifugal pump units covered by this specification shall be the electric-motor-driven, automatic, wet-pit-type. Each pump shall be furnished with a vertical shaft electric motor, a pump-mounting plate, a pump support pipe, a discharge pipe, and a lubrication system. Pumps shall be equipped with an automatic, liquid level control, and a starting switch. The unit shall be complete and ready for operation when placed in the sump, properly alined, and connected to the associated piping and electric power supply. When specified (see 6.2), a duplex pump installation shall be furnished.

3.1.1 Service conditions.

3.1.1.1 Ambient temperature. Unless otherwise specified (see 6.2), the pump shall be able to operate in an ambient temperature of up to 100 degrees Fahrenheit (°F).

3.1.1.2 Material pumped. Unless otherwise specified (see 6.2), the pump unit shall be suitable for pumping drainage or waste water at a maximum temperature of 150°F at sea level.

3.1.1.3 Installation. The pump unit shall be suitable for insertion into sumps, as specified (see 6.1 and 6.2). Plan dimensions of the pump cover plate and the pump length requirements shall be indicated in the invitation for bids in accordance with one of the following options:

a. Depth and plan dimensions of the sump (see 6.1.1).
b. Plan dimensions of the cover plate and the length of the pump from the cover plate to the bottom of the pump suction inlet (see 6.1.2 and 6.1.3).

The length of the pump, determined by the dimension of the pump from the sump cover plate to the bottom of the suction inlet, shall be such as to provide a clearance of not less than one discharge pipe diameter nor more than 12 inches from the floor of the sump.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Design. Pump unit design shall meet the criteria of the Centrifugal Pump Section of the HI Standards and shall be in accordance with the service condition requirements of 3.1.1. The equipment shall be designed to withstand strains, jars, vibrations, and other conditions incidental to shipping, storage, and installation, and to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

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3.4 Standard commercial product. The pump unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the pump unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically specified.

3.5.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. Where such contact cannot be avoided, joints between dissimilar metal shall be separated by an insulating material.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Performance. Each pump unit shall deliver, at rated speed, not less than the specified gpm against the specified discharge head (see 6.2). The liquid level shall be not more than 1 foot above the datum elevation of the pump for group I pumps, and between a minimum of 1 foot and a maximum of 3 feet for group II pumps. The datum elevation shall be taken at the level of the entrance eye of the impeller. The discharge head includes the friction head of the system piping external to the pump unit and the static head measured from the bottom of the cover plate to the highest point in the system. The manufacturer shall allow for the difference in elevation, and for friction losses in the discharge piping. Ratings shall be based on the capacity to pump clear fresh water at a temperature of 68°F at sea level.

3.8 Construction. The pump unit shall be complete so that when installed and connected to the specified source of electricity (see 3.9.1), sump (see 3.1.1.3), and piping system, it can be used to accomplish the operation for which it was designed. The pump shall be connected to the motor by a vertical

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shaft with a flexible coupling, or be a thrust-carrying solid shaft motor with a rigid coupling, or the connection may be made by the use of a thrust-carrying hollow shaft motor.

3.8.1 Pump casing. Unless otherwise specified (see 6.2), the pump casing shall be made of close-grained cast iron, and shall be designed to facilitate inspection and removal of the impeller. Unless otherwise specified (see 6.2), the pump intake shall be provided with a cast iron strainer or an easily removable and cleanable heavy screen of suitable size.

3.8.2 Impeller. Unless otherwise specified (see 6.2), the impeller shall be cast iron. The impeller shall have ample water passages, be mechanically balanced, and be keyed and locked or otherwise securely fastened to the pump shaft to prevent backing-off or loosening with change of rotation.

3.8.3 Pump shaft. Unless otherwise specified (see 6.2), pump shafts shall be high carbon or alloy steel and shall be totally enclosed. The mechanical properties and diameter of the shaft shall be such as to ensure that neither whip, deflection, nor vibration will be of such magnitude under normal operating condition to impose loads. Means shall be provided on the part of the shaft extending above the sump cover plate for adjustment of the clearance between the impeller and the inner surfaces of the volute section.

3.8.4 Suspension pipe. Unless otherwise specified (see 6.2), the suspension pipe or supporting column shall be of heavy wrought steel tubing or cast iron and shall have ample strength and rigidity. Suspension pipe flanges shall be suitably machined or doweled to ensure proper alinement of the pump and shaft wherever the pipe is assembled and reassembled in the field. As an alternative type of construction, group I pump units may have the suspension pipe attached by threaded male and female connections to the pump cover plate or motor support and the pump casing in accordance with the manufacturer's standard practice.

3.8.5 Discharge pipe. Unless otherwise specified (see 6.2), the pump discharge pipe shall be of wrought steel having a wall thickness not less than that specified in ASTM A120 for schedule 40 pipe. The discharge pipe shall be gastight through the sump cover plate. Unless otherwise specified (see 6.2), the discharge end of the pipe shall terminate in an internally threaded or flanged connection of pipe size in accordance with the manufacturer's standard practice.

3.8.6 Sump cover plate. The sump cover plate shall be cast iron or steel and shall be of adequate strength to support the weight of both the pump(s) and maintenance personnel required for normal service without visible distortion. When pumps are to be installed in concrete pits, a grouting frame shall be included with the sump cover plate (see 6.2).

3.8.7 Bearings. One or more radial ball bearings designed and located to carry mechanical and hydraulic thrust loads or rotating components shall be furnished. Thrust bearings shall be designed for not less than 4,000 life

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hours. Thrust bearings shall be located above the pump-mounting plate or cover and fitted with a moisture-proof seal. Intermediate shaft and lower bearings shall be of the sleeve type. Unless otherwise specified (see 6.2), sleeve type bearings shall be heavy-duty bronze, carbon, or molded plastic-lined. When heavy-duty bronze bearings are furnished, appropriate piping and fittings shall be provided to permit individual lubrication of the intermediate and lower bearings from above the sump cover plate.

3.8.8 Motor stand. The motor shall be mounted on a steel or cast iron stand of substantial construction to ensure correct alinement of motor and pump.

3.9 Components and accessories.

3.9.1 Electric motors. Electric motors shall be the vertical type and shall be rated and constructed in accordance with the applicable provisions of NEMA MG-1. Motors shall be general purposed, continuous-duty with a temperature-rise rating of 100°F (or a rating of 100°F ambient). The synchronous speed rating of motors shall not exceed 1,800 rpm for 60 Hertz (Hz) operation and 1,500 rpm for 50 Hz operation. Motors shall be furnished with drip-proof or NEMA type II, weather-protected enclosures as specified (see 6.2). Voltage, phase, and frequency ratings shall be as specified (see 6.2). Provisions shall be made within the motor to prevent overloading at any point on the pump performance curve in accordance with NFPA 70. The horsepower (hp) rating of each pump motor of the duplex pump unit shall be such as to ensure conformance with the following requirements:

a. The motor hp rating shall not be exceeded when the pump is delivering the volume of clear fresh water specified in 3.7 and 4.4.3 against the total dynamic head of the intended installation.
b. When the pump delivers the specified volume of clear fresh water against the total dynamic head of the intended installation, the hp obtained by multiplying the rated motor hp by the service factor shall not be exceeded if the static head is decreased by the length of the pump -1 foot. The condition will occur if the fluid level in sump rises to the cover plate of the pump (see 6.1.4).

3.9.2 Motor starter. Unless otherwise specified (see 6.2), the across-the-line starter shall include thermal overload protection with a hand-off-automatic switch in its cover, and NEMA type 4 enclosure as defined and shall be furnished with capacity and electrical characteristics suitable for the motor according to NFPA 70.

3.9.3 Automatic control(s). All automatic control(s) and float switches shall bear the UL listing mark and shall be listed in the UL Electrical Construction Materials List. Controls shall have the required capacities and electrical characteristics to operate the pump unit. Control circuits shall not exceed 120V ac. Transformer and relay equipment shall be provided for automatic control(s) that operate below 120V.

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3.9.3.1 Float switch. Unless otherwise specified (see 6.1.5 and 6.2), the float switch shall be extra heavy duty, and shall be particularly adapted to float operations. The switch shall be enclosed in a NEMA type 1 or 4 enclosure, as specified (see 6.2).

3.9.3.2 Float. Unless otherwise specified (see 6.2), the switch shall be activated by a copper or plastic float through a noncorrosive float rod, guided above and below the cover plate. The float rod shall be provided with adjustable stops.

3.9.3.3 Alternator controller. When a duplex pump installation is required (see 3.1), a suitable automatic (mechanical or electrical) alternator controller shall be provided, as specified (see 6.1.5 and 6.2). The alternator controller shall be designed for control of alternate and simultaneous operation of the pumps, as required to handle the load.

3.9.3.4 High-water alarm. When specified (see 6.1.5 and 6.2), a high-water alarm complete with actuating-mechanism shall be provided for operation on an electrical circuit other than the pump motor circuit. In the event of pump failure or an inflow greater than the capacity of the pump unit, the electric circuit shall close and operate an alarm bell and warning light, complete with silencing relay.

3.9.4 Wiring. Wiring shall be in accordance with NFPA 70. Provisions shall be made to ground all electrical equipment.

3.10 Safety. All mechanical or electrical parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded in accordance with OSHA regulations, part 1910.219.

3.10.1 Noise. The pumping assembly shall comply with occupational noise exposure requirements of OSHA regulation, part 1926.52 in effect at time of manufacture.

3.11 Fungus resistance. When specified (see 6.2), electrical connections, including terminal and circuit connections, components, and circuit elements shall be coated with varnish conforming to MIL-V-173 except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.12 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall

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be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.13 Treatment and painting. Unless otherwise specified (see 6.2), the pump unit shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the pump unit other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.14 Identification plates. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates.

3.15 Instruction plates. When any special or important procedure is to be followed in operating or servicing the equipment, the unit shall be equipped with an instruction plate, suitably located, describing such procedure. Plates shall be of copper- or zinc-base alloy, or aluminum, adequately secured to the unit.

3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

3.16.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.4 Rivet connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for

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the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.5 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one pump unit when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be a standard production item from the supplier's current inventory provided the unit meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one pump unit. All pump units offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 4.0 percent defective.

4.4 Examination. Each pump unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. Failure of a unit to pass any test shall be cause for rejection of the unit. Certified test reports from an independent laboratory approved by the contracting officer will be accepted as evidence that the unit has passed the required tests. Tests shall be conducted as outlined in the reference documents except as otherwise specified herein.

4.5.1 Functional test. The pump unit shall be operated as required to verify that all mechanical and electrical components will function in a manner that ensures compliance with the requirements of this specification when the equipment is installed at this intended location. Manual activation of float-actuated controls may be employed to simulate fluctuations of the liquid level.

4.5.2 Performance test. The pump unit shall be tested to verify compliance with the hydraulic performance requirements of 3.7 and the electrical requirements of 3.9.1. Test methods and procedures shall be in accordance with the Test Code, Centrifugal Pump Section of the HI Standards.

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During the test run a characteristic head-capacity curve shall be plotted within the limits of the operating range imposed by the hp rating of the motor. Brake hp input to the pump at the specified discharge rate and dynamic head shall also be checked and recorded during the test run. If the length of the pump unit furnished under this specification precludes the use of the manufacturer's available testing facility, a shorter support pipe, pump shaft, discharge pipe, and float mechanisms may be substituted for testing purposes, provided that other requirements for the first article sample are met. Failure of the pump unit to meet any of the performance requirements of this specification shall be cause for rejection.

TABLE I. Classification of defects.

Classification	Defects	Requirement Paragraph
Major:		
101	Pump unit not complete.	1.1, 3.1
102	Material not as specified.	3.5, 3.5.1
103	Component parts not new or interchangeable.	3.6
104	Pump casing not as specified.	3.8.1
105	Impeller not proper type and not as specified.	3.8.2
106	Pump shaft not as specified.	3.8.3
107	Suspension pipe not properly aligned.	3.8.4
108	Discharge pipe not as specified.	3.8.5
109	Sump cover plate not as specified.	3.8.6
110	Bearings not as specified.	3.8.7
111	Electric motor not of the type specified or rating required.	3.9.1
112	Motor starter not as specified.	3.9.2
113	Automatic controls not as specified with specified type of enclosure.	3.9.3, 3.9.3.1, 3.9.3.2
114	Alternator controller not furnished as specified.	3.9.3.3
115	High water alarm not furnished when specified.	3.9.3.4
116	Improper wiring.	3.9.4
117	Not treated for fungus resistance, when required.	3.11
118	Pump not lubricated and lubrication fittings not as specified.	3.12

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TABLE I. Classification of defects. (Continued)

Classification	Defects	Requirement Paragraph
Minor:		
201	Treatment and painting not as specified.	3.13
202	Equipment not marked as specified or not furnished with instruction plates when required.	3.14, 3.15
203	Poor workmanship.	3.16

4.6 Packaging inspection. The preservation, packing, and marking of the pump unit shall be inspected to verify conformance to the requirements in section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. The pumps shall be preserved, packed, and marked in accordance with MIL-P-10603. The level of preservation and packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. Pump units are intended for use in metal plating shops, in concrete block curing plants, for various construction jobs, and wherever water gathers (building basements, pits, etc.) and needs to be emptied at a higher level. Such pumps are not intended for conveyance of solids in the waste water.

6.1.1 Sump volume. With float switch control, sump volume should be sized to allow no more than three of four starts per hour per pump. The minimum pit volume in gallons should equal or exceed twice the maximum capacity in gpm to be pumped.

6.1.2 Sump cover plate. Cover plates vary in diameter from 15 to 78 inches. Manufacturers should be consulted for the availability of a suitable cover plate.

6.1.3 Pump length. The minimum pump length usually is 2 feet. The length increases in increments of 6 inches to a maximum of 20 feet.

6.1.4 Capacity. Some impellers are so designed that, as capacity increases with reduction in head, the hp input requirements of the pump also increases. The total head should therefore be determined as accurately as possible to eliminate possible overloading of the motor by specifying a head that is greater than the actual head against which the pump will be required to operate.

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6.1.5 Control and float switch enclosures. When installed on the cover plate or motor mount, enclosures for control and float switches should be of the same NEMA type enclosure.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Required capacity of each pump delivery in gpm and total dynamic head in feet of water (see 1.2 and 3.7).
c. When a duplex pump installation is to be furnished (see 3.1).
d. Service conditions of water and ambient temperature, if different (see 3.1.1.1 and 3.1.1.2).
e. Depth and plan dimensions of the sump or plan dimensions of thecover plate and the length of the pump (see 3.1.1.3, 6.1.1, 6.1.2, and 6.1.3).
f. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
g. Pump casing, if different (see 3.8.1).
h. When strainer is not required (see 3.8.1).
i. Pump impeller, if different (see 3.8.2).
j. Pump shaft, if different (see 3.8.3).
k. Pump suspension pipe, if different (see 3.8.5).
l. Pump discharge pipe, if different (see 3.8.5).
m. Connection at end of discharge pipe, if different (see 3.8.5).
n. When grouting frame is required (see 3.8.6).
o. Type of support sleeve bearings, if different (see 3.8.7).
p. Type of motor enclosure, required (see 3.9.1).
q. Voltage, phase, and frequency of electric power at site of installation (see 3.9.1).
r. Motor starter, if different (see 3.9.2).
s. Float switch, if different, and type of enclosure (see 3.9.3.1).
t. Type of float, if different (see 3.9.3.2).
u. When an alternator controller is required (see 3.9.3.3).
v. When high-water alarm is required (see 3.9.3.4).
w. When fungus resistance is required (see 3.11).
x. When lifting attachments are required (see 3.14).
y. Treating and painting, if different (see 3.15).
z. Marking is other than specified (see 5.1).
aa. Level of preservation and packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (keyword) listing.

Centrifugal
Electric-motor-driven
Pump unit
Sump
Vertical

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - YD

Project 4320-N256

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-21214C(YD)	PUMP UNIT, CENTRIFUGAL, VERTICAL SUMP, ELECTRIC MOTOR DRIVEN, AUTOMATIC, WET-PIT-TYPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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